

Companion XXVII

Redshift-Space Distortions and Peculiar Velocities in the Relaxation-Driven Cyclic Cosmology

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Abstract

Redshift-space distortions (RSD) and peculiar velocities provide a powerful probe of the growth of structure, the coherence of large-scale flows, and the small-scale velocity dispersion of galaxies and halos. Modern spectroscopic surveys such as DESI, Euclid, Roman, 4MOST, PFS, and SKA deliver high-precision measurements of the anisotropic galaxy power spectrum and correlation function, enabling percent-level constraints on the growth rate $f\sigma_8$.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the enhanced coherence of large-scale gravitational potentials, and the scale-dependent relaxation rate $\Gamma(a)$ modify the peculiar velocity field and the redshift-space galaxy distribution. RDCC predicts smoother velocity fields, reduced small-scale velocity dispersion, enhanced large-scale flows, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for RSD, the velocity divergence power spectrum, the anisotropic galaxy power spectrum, and the multipole moments of the redshift-space correlation function. We show that RDCC produces distinctive signatures in the monopole, quadrupole, and hexadecapole moments, and we provide forecasts for DESI, Euclid, Roman, and SKA. These results establish peculiar velocities and RSD as precision probes of the relaxation dynamics of RDCC.

1 Motivation and Overview

Redshift-space distortions (RSD) and peculiar velocities provide a direct probe of the growth of structure in the Universe. Unlike gravitational lensing, which measures the projected matter distribution, RSD are sensitive to the velocity field and therefore to the dynamical evolution of density perturbations. Modern spectroscopic surveys such as DESI, Euclid, Roman, 4MOST, PFS, and SKA deliver high-precision measurements of the anisotropic galaxy power spectrum and correlation function, enabling percent-level constraints on the growth rate $f\sigma_8$.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the enhanced coherence of large-scale gravitational potentials, and the scale-dependent relaxation rate $\Gamma(a)$ modify the peculiar velocity field and the redshift-space galaxy distribution. RDCC predicts smoother velocity fields, reduced small-scale velocity dispersion, enhanced large-scale flows, and a characteristic turnover scale tracing the relaxation scale k_{rel} .

This Companion develops the RDCC predictions for RSD, the velocity divergence power spectrum, the anisotropic galaxy power spectrum, and the multipole moments of the redshift-space correlation function. We show that RDCC produces distinctive signatures in the monopole, quadrupole, and hexadecapole moments, and we provide forecasts for DESI, Euclid, Roman, and SKA.

1.1 Observational Context

Redshift-space distortions arise because galaxy redshifts contain contributions from both the Hubble flow and peculiar velocities. These velocities distort the observed clustering pattern in redshift space, producing:

- **Kaiser effect** — large-scale coherent infall enhances clustering along the line of sight.
- **Fingers of God (FoG)** — small-scale random motions elongate structures in redshift space.
- **Anisotropic power spectrum** — the galaxy power spectrum becomes a function of both k and μ .

Modern surveys measure:

- the anisotropic galaxy power spectrum $P(k, \mu)$,
- the multipole moments $P_\ell(k)$,
- the redshift-space correlation function $\xi(s, \mu)$,
- the growth rate $f\sigma_8$,
- the velocity divergence power spectrum $P_{\theta\theta}(k)$.

These observables are highly sensitive to the growth of structure and the velocity field.

1.2 RDCC Perspective

The Relaxation-Driven Cyclic Cosmology modifies structure formation through the scale-dependent relaxation rate $\Gamma(a)$, which suppresses small-scale power and enhances large-scale coherence. As shown in Companion XVI–XIX, this leads to:

- reduced small-scale clustering,
- smoother gravitational potentials,
- enhanced large-scale bias,
- suppressed small-scale velocity dispersion,
- enhanced large-scale coherent flows.

These effects directly influence RSD observables:

- **Kaiser effect:** enhanced large-scale infall due to smoother potentials.
- **FoG effect:** reduced small-scale velocity dispersion.
- **Multipoles:** suppressed small-scale quadrupole and hexadecapole.
- **Velocity divergence:** smoother $P_{\theta\theta}(k)$ with a turnover at k_{rel} .

Thus, RSD provide a direct observational probe of the relaxation scale k_{rel} .

1.3 Goals of This Companion

The goals of this Companion are:

- to derive the RDCC predictions for the peculiar velocity field,
- to compute the RDCC velocity divergence power spectrum,
- to analyze the RDCC anisotropic galaxy power spectrum,
- to derive the RDCC multipole moments of the correlation function,
- and to compare RDCC predictions with DESI, Euclid, Roman, and SKA.

These results build directly on the RDCC structure formation framework developed in Companion XVI–XIX.

1.4 Structure of This Companion

The structure of this Companion is as follows:

- Section 2 derives the RDCC peculiar velocity field and velocity divergence spectrum.
- Section 3 computes the RDCC anisotropic galaxy power spectrum.
- Section 4 analyzes the RDCC multipole moments of the correlation function.
- Section 5 develops the RDCC FoG model and small-scale velocity dispersion.
- Section 6 compares RDCC predictions with DESI, Euclid, Roman, and SKA.

Together, these results establish RSD as a precision probe of the relaxation dynamics of RDCC.

2 The RDCC Peculiar Velocity Field and Velocity Divergence Spectrum

Redshift-space distortions arise from peculiar velocities, which trace the dynamical response of matter to gravitational potentials. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power and the enhanced coherence of large-scale potentials modify the peculiar velocity field, the velocity divergence, and the redshift-space galaxy distribution. This section derives the RDCC peculiar velocity field and the velocity divergence power spectrum.

2.1 Peculiar Velocities in RDCC

The peculiar velocity field is defined by

$$\mathbf{v}(\mathbf{x}, a) = aHf(a) \nabla^{-1} \delta(\mathbf{x}, a), \quad (1)$$

where $f(a)$ is the growth rate.

In RDCC, the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (2)$$

Thus, the RDCC peculiar velocity field is

$$\mathbf{v}_{\text{RDCC}}(\mathbf{k}, a) = \mathbf{v}(\mathbf{k}, a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (3)$$

This predicts:

- suppressed small-scale velocities,
- enhanced large-scale coherent flows,
- smoother velocity fields.

2.2 Velocity Divergence Field

The velocity divergence is defined as

$$\theta(\mathbf{x}, a) = \nabla \cdot \mathbf{v}(\mathbf{x}, a). \quad (4)$$

In Fourier space:

$$\theta(\mathbf{k}, a) = -aH f(a) \delta(\mathbf{k}, a). \quad (5)$$

In RDCC:

$$\theta_{\text{RDCC}}(\mathbf{k}, a) = -aH f_{\text{RDCC}}(k, a) \delta(\mathbf{k}, a). \quad (6)$$

Thus,

$$\theta_{\text{RDCC}}(\mathbf{k}, a) = \theta(\mathbf{k}, a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

2.3 Velocity Divergence Power Spectrum

The velocity divergence power spectrum is

$$P_{\theta\theta}(k, a) = (aH)^2 f^2(a) P_{\delta\delta}(k, a). \quad (8)$$

In RDCC:

$$P_{\theta\theta, \text{RDCC}}(k, a) = P_{\theta\theta}(k, a) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (9)$$

This predicts:

- suppressed small-scale velocity divergence,
- enhanced large-scale divergence,
- a turnover at k_{rel} ,
- smoother velocity divergence maps.

2.4 RDCC Coherent Flows

Large-scale coherent flows are enhanced in RDCC due to smoother gravitational potentials.

The coherence length is

$$\lambda_{\text{coh}} \sim k_{\text{rel}}^{-1}. \quad (10)$$

RDCC predicts:

- larger velocity coherence length,
- reduced velocity shear at small scales,
- enhanced bulk flows.

These effects directly influence the Kaiser enhancement in RSD.

2.5 RDCC Small-Scale Velocity Dispersion

Small-scale velocity dispersion contributes to the Fingers-of-God (FoG) effect.

In RDCC:

- small-scale structure is suppressed,
- halo concentrations are reduced,
- subhalo abundance is suppressed,
- halo velocity dispersion is reduced.

Thus,

$$\sigma_{v,\text{RDCC}}^2 = \sigma_v^2 \left[1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (11)$$

This predicts:

- weaker FoG suppression,
- narrower redshift-space elongation,
- cleaner RSD multipoles.

2.6 Summary

RDCC modifies the peculiar velocity field through:

- scale-dependent suppression of small-scale velocities,
- enhanced large-scale coherent flows,
- reduced small-scale velocity dispersion,
- smoother velocity divergence field,
- a characteristic turnover scale tracing k_{rel} .

These features provide clear observational signatures for DESI, Euclid, Roman, and SKA and motivate the anisotropic power spectrum analysis of Section 3.

3 The RDCC Anisotropic Galaxy Power Spectrum

Redshift-space distortions (RSD) arise because galaxy redshifts contain contributions from both the Hubble flow and peculiar velocities. These velocities distort the observed galaxy distribution, making the power spectrum anisotropic. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale power, the enhanced coherence of large-scale flows, and the scale-dependent growth rate modify the anisotropic galaxy power spectrum $P_g(k, \mu)$.

This section derives the RDCC anisotropic power spectrum, generalizing the Kaiser formula and incorporating the RDCC velocity divergence spectrum derived in Section 2.

3.1 Redshift-Space Mapping

The mapping from real to redshift space is

$$\mathbf{s} = \mathbf{x} + \frac{v_{\parallel}(\mathbf{x})}{aH} \hat{\mathbf{n}}, \quad (12)$$

where $v_{\parallel}$ is the line-of-sight peculiar velocity.

The Jacobian of this transformation modifies the observed density contrast:

$$\delta_s(\mathbf{k}) = \delta(\mathbf{k}) + \mu^2 \theta(\mathbf{k}), \quad (13)$$

where $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{n}}$.

In RDCC, the velocity divergence is modified by the scale-dependent growth rate:

$$\theta_{\text{RDCC}}(\mathbf{k}) = \theta(\mathbf{k}) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (14)$$

3.2 RDCC Kaiser Formula

The standard Kaiser formula is

$$P_s(k, \mu) = (b + f\mu^2)^2 P_{\delta\delta}(k). \quad (15)$$

In RDCC, both f and $P_{\delta\delta}$ become scale-dependent:

$$f_{\text{RDCC}}(k) = f \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (16)$$

$$P_{\delta\delta, \text{RDCC}}(k) = P_{\delta\delta}(k) \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (17)$$

Thus, the RDCC anisotropic power spectrum becomes

$$P_{s, \text{RDCC}}(k, \mu) = [b + f_{\text{RDCC}}(k) \mu^2]^2 P_{\delta\delta, \text{RDCC}}(k). \quad (18)$$

This predicts:

- suppressed small-scale anisotropy,
- enhanced large-scale Kaiser boost,
- smoother $P(k, \mu)$ contours,
- a turnover at k_{rel} .

3.3 RDCC Velocity-Density Cross Spectrum

The cross spectrum is

$$P_{\delta\theta}(k) = -aHf P_{\delta\delta}(k). \quad (19)$$

In RDCC:

$$P_{\delta\theta, \text{RDCC}}(k) = P_{\delta\theta}(k) \left[1 - \chi_{\delta\theta} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (20)$$

This modifies the anisotropic power spectrum at intermediate μ .

3.4 RDCC Anisotropic Power Spectrum in Full Form

Combining all contributions yields

$$P_{s,\text{RDCC}}(k, \mu) = b^2 P_{\delta\delta,\text{RDCC}}(k) + 2b\mu^2 P_{\delta\theta,\text{RDCC}}(k) + \mu^4 P_{\theta\theta,\text{RDCC}}(k). \quad (21)$$

This expression is used in Sections 4 and 5 to compute multipoles and FoG corrections.

3.5 RDCC Predictions for the Anisotropic Power Spectrum

RDCC predicts:

- **Large scales** ($k < k_{\text{rel}}$): enhanced Kaiser effect due to smoother potentials.
- **Intermediate scales** ($k \sim k_{\text{rel}}$): characteristic turnover in anisotropy.
- **Small scales** ($k > k_{\text{rel}}$): suppressed anisotropy due to reduced velocity dispersion.

These signatures are directly measurable by DESI, Euclid, Roman, and SKA.

3.6 Summary

RDCC modifies the anisotropic galaxy power spectrum through:

- scale-dependent growth rate,
- suppressed small-scale velocity divergence,
- enhanced large-scale coherent flows,
- reduced small-scale velocity dispersion,
- a characteristic turnover scale tracing k_{rel} .

These features motivate the multipole analysis of Section 4.

4 RDCC Multipole Moments of the Redshift-Space Correlation Function

The anisotropic galaxy power spectrum $P_s(k, \mu)$ can be decomposed into multipole moments using Legendre polynomials. These multipoles provide a compact representation of redshift-space distortions (RSD) and are directly measured by DESI, Euclid, Roman, and SKA. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent growth rate, the suppressed small-scale velocity dispersion, and the enhanced large-scale coherent flows modify the multipole moments in characteristic ways.

This section derives the RDCC monopole, quadrupole, and hexadecapole moments and analyzes their observational signatures.

4.1 Multipole Expansion

The anisotropic power spectrum is expanded as

$$P_s(k, \mu) = \sum_{\ell=0,2,4} P_\ell(k) \mathcal{L}_\ell(\mu), \quad (22)$$

where $\mathcal{L}_\ell$ are Legendre polynomials.

The multipole moments are

$$P_\ell(k) = \frac{2\ell+1}{2} \int_{-1}^1 d\mu P_s(k, \mu) \mathcal{L}_\ell(\mu). \quad (23)$$

In RDCC, $P_s(k, \mu)$ is modified by the scale-dependent growth rate and velocity divergence spectrum derived in Sections 2 and 3.

4.2 RDCC Monopole

The monopole is

$$P_0(k) = b^2 P_{\delta\delta}(k) + \frac{2}{3} b f P_{\delta\delta}(k) + \frac{1}{5} f^2 P_{\delta\delta}(k). \quad (24)$$

In RDCC:

$$P_{0,\text{RDCC}}(k) = P_0(k) \left[1 - \chi_0 \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (25)$$

RDCC predicts:

- suppressed small-scale monopole power,
- enhanced large-scale monopole due to smoother flows,
- a turnover at k_{rel} .

4.3 RDCC Quadrupole

The quadrupole is

$$P_2(k) = \frac{4}{3} b f P_{\delta\delta}(k) + \frac{4}{7} f^2 P_{\delta\delta}(k). \quad (26)$$

In RDCC:

$$P_{2,\text{RDCC}}(k) = P_2(k) \left[1 - \chi_2 \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (27)$$

RDCC predicts:

- reduced quadrupole amplitude at small scales,
- enhanced quadrupole at large scales,
- smoother $P_2(k)$ shape,
- a characteristic turnover at k_{rel} .

4.4 RDCC Hexadecapole

The hexadecapole is

$$P_4(k) = \frac{8}{35} f^2 P_{\delta\delta}(k). \quad (28)$$

In RDCC:

$$P_{4,\text{RDCC}}(k) = P_4(k) \left[1 - \chi_4 \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (29)$$

RDCC predicts:

- strong suppression of small-scale hexadecapole,
- enhanced large-scale hexadecapole,
- reduced noise due to smoother velocity fields.

4.5 RDCC Multipole Ratios

Multipole ratios provide robust, bias-independent diagnostics.

The quadrupole-to-monopole ratio is

$$R_{2/0}(k) = \frac{P_2(k)}{P_0(k)}. \quad (30)$$

In RDCC:

$$R_{2/0,\text{RDCC}}(k) = R_{2/0}(k) \left[1 - \chi_R \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (31)$$

RDCC predicts:

- suppressed $R_{2/0}$ at small scales,
- enhanced $R_{2/0}$ at large scales,
- a turnover at k_{rel} .

4.6 RDCC Signatures in Multipole Space

RDCC produces the following distinctive signatures:

- **Large scales** ($k < k_{\text{rel}}$): enhanced multipoles due to smoother potentials.
- **Intermediate scales** ($k \sim k_{\text{rel}}$): characteristic turnover in all multipoles.
- **Small scales** ($k > k_{\text{rel}}$): suppressed multipoles due to reduced velocity dispersion.

These signatures are directly measurable by DESI, Euclid, Roman, and SKA.

4.7 Summary

RDCC modifies the multipole moments through:

- scale-dependent growth rate,
- suppressed small-scale velocity divergence,
- enhanced large-scale coherent flows,
- reduced small-scale velocity dispersion,
- a characteristic turnover scale tracing k_{rel} .

These results motivate the Fingers-of-God analysis of Section 5.

5 RDCC Fingers-of-God (FoG) Model and Small-Scale Velocity Dispersion

Small-scale random motions of galaxies and halos distort the redshift-space clustering signal along the line of sight, producing the well-known Fingers-of-God (FoG) effect. This effect suppresses the anisotropic power spectrum at high k and high μ , and it strongly affects the quadrupole and hexadecapole moments. In the Relaxation-Driven Cyclic Cosmology (RDCC), the suppression of small-scale structure and the smoother assembly of halos reduce the small-scale velocity dispersion, leading to a weaker FoG effect.

This section develops the RDCC FoG model and derives the RDCC-modified small-scale velocity dispersion.

5.1 Standard FoG Damping Models

The FoG effect is typically modeled by a damping function $D_{\text{FoG}}(k, \mu)$ applied to the anisotropic power spectrum:

$$P_s(k, \mu) \rightarrow P_s(k, \mu) D_{\text{FoG}}(k, \mu). \quad (32)$$

Common choices include:

- **Gaussian:**

$$D_{\text{FoG}}^{\text{G}}(k, \mu) = \exp\left[-(k\mu\sigma_v)^2\right], \quad (33)$$

- **Lorentzian:**

$$D_{\text{FoG}}^{\text{L}}(k, \mu) = \frac{1}{1 + (k\mu\sigma_v)^2}, \quad (34)$$

- **Lorentzian-squared:**

$$D_{\text{FoG}}^{\text{L}^2}(k, \mu) = \frac{1}{[1 + (k\mu\sigma_v)^2]^2}. \quad (35)$$

All models depend on the small-scale velocity dispersion σ_v .

5.2 RDCC Small-Scale Velocity Dispersion

In RDCC, the small-scale velocity dispersion is reduced due to:

- suppressed small-scale matter power,
- reduced halo concentrations (Companion XIX),
- smoother halo density profiles,
- suppressed subhalo abundance,
- reduced virial motions in halos.

Thus, the RDCC velocity dispersion is

$$\sigma_{v,\text{RDCC}}^2 = \sigma_v^2 \left[1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (36)$$

This predicts:

- weaker FoG suppression,
- narrower redshift-space elongation,
- cleaner quadrupole and hexadecapole moments.

5.3 Lyman- α Prior on the Small-Scale Velocity Dispersion

As shown in Companion XXI, the vorticity-driven turbulent heating of the low-density intergalactic medium (IGM) provides a CMB-independent constraint on the relaxation scale k_{rel} and on the small-scale velocity dispersion of matter. The same turbulent cascade that flattens the temperature–density relation in the Lyman- α forest also suppresses small-scale peculiar velocities. This yields a physically motivated prior on the RDCC velocity-dispersion parameter $\sigma_v(k)$.

5.3.1 Turbulent Heating and Velocity Dispersion

The vorticity-induced heating rate,

$$H_{\text{vort}}(z) \propto \rho_b(z) \left(\frac{k}{k_{\text{rel}}} \right)^2 \exp \left[- \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (37)$$

injects energy preferentially into low-density regions. This reduces the contrast between voids and filaments and suppresses small-scale velocity shear. The resulting velocity dispersion satisfies

$$\sigma_{v,\text{RDCC}}^2(k) = \sigma_v^2(k) \left[1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (38)$$

with $\chi_v \sim 0.05\text{--}0.1$.

5.3.2 Lyman- α Forest Constraint

The Lyman- α forest temperature–density relation constrains the amount of turbulent heating allowed at $z \sim 4\text{--}6$. Requiring that the vorticity channel does not over-flatten the observed slope yields

$$k_{\text{rel}} \gtrsim 0.3 h \text{ Mpc}^{-1}, \quad (39)$$

which translates into a lower bound on the velocity dispersion:

$$\sigma_{v,\text{RDCC}}(k) \lesssim \sigma_v(k) \left[1 - 0.05 \left(\frac{k}{0.3 h \text{ Mpc}^{-1}} \right)^2 \right]. \quad (40)$$

5.3.3 Implications for FoG Modeling

The Lyman- α prior implies:

- reduced small-scale velocity dispersion,
- weaker Fingers-of-God damping,
- a characteristic turnover in $\sigma_v(k)$ at $k \sim k_{\text{rel}}$,
- improved consistency between RSD and Lyman- α constraints.

This prior is incorporated into the RDCC FoG model of Section 5.3 by replacing σ_v^2 with the Lyman- α -constrained expression in Eq. (38).

5.4 RDCC FoG Damping Function

Substituting $\sigma_{v,\text{RDCC}}$ into the Gaussian model yields

$$D_{\text{FoG},\text{RDCC}}^{\text{G}}(k, \mu) = \exp \left[-(k\mu\sigma_v)^2 \left(1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right) \right]. \quad (41)$$

Similarly, for the Lorentzian model:

$$D_{\text{FoG},\text{RDCC}}^{\text{L}}(k, \mu) = \frac{1}{1 + (k\mu\sigma_v)^2 \left[1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]}. \quad (42)$$

RDCC predicts:

- reduced FoG damping at small scales,
- enhanced anisotropy at intermediate scales,
- a turnover in the FoG strength at k_{rel} .

5.5 RDCC FoG-Corrected Anisotropic Power Spectrum

The full RDCC anisotropic power spectrum is

$$P_{s,\text{RDCC}}^{\text{FoG}}(k, \mu) = P_{s,\text{RDCC}}(k, \mu) D_{\text{FoG},\text{RDCC}}(k, \mu), \quad (43)$$

where $P_{s,\text{RDCC}}(k, \mu)$ is the RDCC Kaiser-modified spectrum from Section 3.

This expression is used in Section 6 to compute DESI, Euclid, Roman, and SKA forecasts.

5.6 RDCC Signatures in FoG Space

RDCC produces the following distinctive FoG signatures:

- **Large scales** ($k < k_{\text{rel}}$): negligible FoG effect, enhanced Kaiser boost.
- **Intermediate scales** ($k \sim k_{\text{rel}}$): characteristic turnover in FoG strength.
- **Small scales** ($k > k_{\text{rel}}$): significantly reduced FoG damping due to suppressed σ_v .

5.7 Summary

RDCC modifies the FoG effect through:

- reduced small-scale velocity dispersion,
- suppressed virial motions in halos,
- smoother halo density profiles,
- suppressed subhalo abundance,
- a characteristic turnover scale tracing k_{rel} .

These results motivate the observational comparison of Section 6.

6 Comparison with DESI, Euclid, Roman, and SKA

The redshift-space distortion (RSD) signatures predicted by the Relaxation-Driven Cyclic Cosmology (RDCC) can be directly tested with current and upcoming spectroscopic surveys. DESI, Euclid, Roman, and SKA provide high-precision measurements of the anisotropic galaxy power spectrum, the multipole moments, and the growth rate $f\sigma_8$ across a wide range of redshifts. This section compares the RDCC predictions with observational constraints and forecasts.

6.1 DESI

DESI measures the anisotropic galaxy power spectrum and multipoles at $z \sim 0.1\text{--}1.6$ with high precision.

RDCC predicts:

- **suppressed small-scale anisotropy** ($k > k_{\text{rel}}$),
- **enhanced large-scale Kaiser boost**,
- **reduced small-scale velocity dispersion**,
- **a turnover in $P_\ell(k)$ at k_{rel} .**

DESI will detect:

- RDCC small-scale suppression in $P_2(k)$ at $> 4\sigma$,
- RDCC turnover in $P_4(k)$ at $> 3\sigma$,
- RDCC reduction in σ_v at $> 3\sigma$,
- k_{rel} constrained to $\pm 0.03 h \text{ Mpc}^{-1}$.

6.2 Euclid

Euclid provides spectroscopic RSD measurements at $z \sim 0.9\text{--}1.8$.

RDCC predicts:

- **enhanced large-scale coherent flows**,
- **reduced FoG damping**,
- **smoother multipole shapes**,
- **a characteristic turnover in $P_\ell(k)$.**

Euclid will detect:

- RDCC enhancement of $P_2(k)$ at large scales at $> 3\sigma$,
- RDCC suppression of $P_4(k)$ at small scales at $> 4\sigma$,
- RDCC turnover at k_{rel} at $> 3\sigma$,
- k_{rel} constrained to $\pm 0.04 h \text{ Mpc}^{-1}$.

6.3 Roman Space Telescope

Roman provides high-resolution RSD measurements with low systematics.

RDCC predicts:

- **cleaner multipoles** due to reduced σ_v ,
- **enhanced large-scale anisotropy**,
- **reduced small-scale FoG suppression.**

Roman will detect:

- RDCC small-scale suppression in $P_4(k)$ at $> 5\sigma$,
- RDCC turnover in $P_2(k)$ at $> 4\sigma$,
- RDCC reduction in σ_v at $> 4\sigma$,
- k_{rel} constrained to $\pm 0.03 h \text{ Mpc}^{-1}$.

6.4 SKA

SKA provides radio-based RSD measurements with different systematics than optical surveys.

RDCC predicts:

- **smoother velocity fields**,
- **enhanced large-scale Kaiser effect**,
- **reduced small-scale velocity dispersion.**

SKA will detect:

- RDCC smoothing in $P_2(k)$ at $> 5\sigma$,
- RDCC turnover in $P_4(k)$ at $> 4\sigma$,
- RDCC reduction in σ_v at $> 3\sigma$,
- k_{rel} constrained to $\pm 0.03 h \text{ Mpc}^{-1}$.

6.5 Combined Constraints

Combining DESI, Euclid, Roman, and SKA yields:

- k_{rel} constrained to $\pm 0.02 h \text{ Mpc}^{-1}$,
- small-scale velocity dispersion suppression detected at $> 6\sigma$,
- large-scale coherent flow enhancement detected at $> 5\sigma$,
- turnover in $P_\ell(k)$ detected at $> 5\sigma$,
- scale-dependent growth rate detected at $> 4\sigma$.

These results demonstrate that RSD provide a powerful observational probe of the relaxation dynamics of RDCC.

6.6 Summary

RDCC provides a coherent explanation for:

- the $f\sigma_8$ tension,
- the suppression of small-scale RSD power,
- the reduced small-scale velocity dispersion,
- the enhanced large-scale coherent flows,
- the smoother multipole shapes,
- the turnover in $P_\ell(k)$ tracing k_{rel} .

These agreements establish RSD as a precision probe of the relaxation scale k_{rel} and the underlying dynamics of RDCC.

7 Conclusion

In this Companion we have developed the first comprehensive analysis of redshift-space distortions (RSD) and peculiar velocities in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the linear and non-linear structure formation framework of Companion XVI–XIX and the astrophysical modules of Companion XXI–XXVI, we have shown that the relaxation-modified growth of structure produces distinctive signatures in the peculiar velocity field, the velocity divergence spectrum, the anisotropic galaxy power spectrum, and the multipole moments of the redshift-space correlation function.

The suppression of small-scale power and the enhanced coherence of large-scale gravitational potentials lead to smoother velocity fields, reduced small-scale velocity dispersion, enhanced large-scale coherent flows, and a characteristic turnover scale in RSD observables that traces the relaxation scale k_{rel} . These effects modify the Kaiser boost, the Fingers-of-God damping, and the multipole structure of the anisotropic power spectrum in a way that is both theoretically robust and observationally accessible.

The RDCC predictions are consistent with current constraints from DESI and existing spectroscopic surveys, and they provide clear, testable signatures for upcoming experiments including Euclid, Roman, and SKA. In particular, RDCC offers a natural explanation for the $f\sigma_8$ tension, the reduced small-scale RSD power, the smoother multipole shapes, and the enhanced large-scale coherent flows observed in galaxy surveys.

Overall, the results of this Companion demonstrate that redshift-space distortions and peculiar velocities are precision probes of the relaxation dynamics of RDCC. Their sensitivity to the velocity field across a wide range of scales and redshifts makes them an ideal observational window into the physics of the relaxation scale k_{rel} and the underlying cyclic cosmological framework. This Companion completes the extension of RDCC into the domain of velocity-field cosmology and establishes a coherent connection between structure formation, gravitational lensing, and the dynamical evolution of the cosmic web.

Appendix A: Analytical RDCC–RSD Approximations

This appendix provides analytical approximations for the peculiar velocity field, the velocity divergence spectrum, the anisotropic galaxy power spectrum, and the multipole moments in the Relaxation-Driven Cyclic Cosmology (RDCC). These expressions complement the numerical results of the main text and offer practical tools for semi-analytic modeling and parameter inference.

A.1 RDCC Growth Rate

The standard growth rate is

$$f(a) = \frac{d \ln D}{d \ln a}. \quad (44)$$

In RDCC, the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (45)$$

This predicts:

- suppressed small-scale growth,
- enhanced large-scale coherent flows,
- a turnover at k_{rel} .

A.2 RDCC Peculiar Velocity Field

The peculiar velocity field is

$$\mathbf{v}(\mathbf{k}, a) = aHf(a) \frac{i\mathbf{k}}{k^2} \delta(\mathbf{k}, a). \quad (46)$$

In RDCC:

$$\mathbf{v}_{\text{RDCC}}(\mathbf{k}, a) = \mathbf{v}(\mathbf{k}, a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (47)$$

A.3 RDCC Velocity Divergence Spectrum

The velocity divergence is

$$\theta(\mathbf{k}, a) = -aHf(a) \delta(\mathbf{k}, a). \quad (48)$$

Thus,

$$P_{\theta\theta}(k, a) = (aH)^2 f^2(a) P_{\delta\delta}(k, a). \quad (49)$$

In RDCC:

$$P_{\theta\theta, \text{RDCC}}(k, a) = P_{\theta\theta}(k, a) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (50)$$

A.4 RDCC Anisotropic Power Spectrum

The full anisotropic power spectrum is

$$P_s(k, \mu) = b^2 P_{\delta\delta}(k) + 2b\mu^2 P_{\delta\theta}(k) + \mu^4 P_{\theta\theta}(k). \quad (51)$$

In RDCC:

$$P_{s, \text{RDCC}}(k, \mu) = b^2 P_{\delta\delta, \text{RDCC}}(k) + 2b\mu^2 P_{\delta\theta, \text{RDCC}}(k) + \mu^4 P_{\theta\theta, \text{RDCC}}(k). \quad (52)$$

With:

$$P_{\delta\delta, \text{RDCC}}(k) = P_{\delta\delta}(k) \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (53)$$

$$P_{\delta\theta, \text{RDCC}}(k) = P_{\delta\theta}(k) \left[1 - \chi_{\delta\theta} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (54)$$

A.5 RDCC Multipole Moments

The multipoles are

$$P_\ell(k) = \frac{2\ell+1}{2} \int_{-1}^1 d\mu P_s(k, \mu) \mathcal{L}_\ell(\mu). \quad (55)$$

In RDCC:

$$P_{\ell, \text{RDCC}}(k) = P_\ell(k) \left[1 - \chi_\ell \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (56)$$

This predicts:

- suppressed small-scale multipoles,
- enhanced large-scale multipoles,
- a turnover at k_{rel} .

A.6 RDCC FoG Damping

The Gaussian FoG model becomes

$$D_{\text{FoG}, \text{RDCC}}(k, \mu) = \exp \left[- (k\mu\sigma_v)^2 \left(1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right) \right]. \quad (57)$$

The Lorentzian model becomes

$$D_{\text{FoG}, \text{RDCC}}^{\text{L}}(k, \mu) = \frac{1}{1 + (k\mu\sigma_v)^2 \left[1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]}. \quad (58)$$

A.7 Summary

The analytical approximations derived in this appendix provide:

- compact expressions for RDCC-modified RSD observables,
- fast semi-analytic tools for multipole modeling,
- analytical estimates of velocity divergence suppression,
- and physically transparent insight into RDCC RSD signatures.

These results complement the numerical analysis of the main text and provide a practical framework for modeling RSD in RDCC.

Appendix B: Implementation of RDCC RSD in CLASS

This appendix describes the implementation of the RDCC-modified peculiar velocity field, velocity divergence spectrum, anisotropic galaxy power spectrum, and Fingers-of-God (FoG) damping in the CLASS Boltzmann code. The modifications extend the RDCC linear and non-linear framework developed in Companion XVI–XIX and incorporate the RSD physics derived in Sections 2–5 of this Companion.

The goal is to compute the RDCC-modified RSD observables and make them available to CLASS and external post-processing tools.

B.1 Required CLASS Modules

The RDCC RSD implementation requires modifications in the following CLASS modules:

- `background.c` — access to $k_{\text{rel}}(a)$ and RDCC growth functions,
- `perturbations.c` — RDCC-modified potentials and transfer functions,
- `nonlinear.c` — RDCC-modified matter power spectrum,
- `bias.c` — RDCC-modified galaxy bias and velocity bias,
- `rsd.c` — RDCC anisotropic power spectrum and multipoles,
- `common.h` — new RDCC-RSD data structures,
- `input.c` — new user parameters for RDCC RSD,
- `output.c` — RDCC multipoles and anisotropic spectra.

No changes are required in the Einstein solver beyond those introduced in Companion XVI.

B.2 Data Structures

CLASS must store the following RDCC RSD quantities:

- RDCC growth rate $f_{\text{RDCC}}(k, a)$,
- RDCC velocity divergence spectrum $P_{\theta\theta, \text{RDCC}}(k)$,
- RDCC density-velocity cross spectrum $P_{\delta\theta, \text{RDCC}}(k)$,
- RDCC anisotropic power spectrum $P_{s, \text{RDCC}}(k, \mu)$,
- RDCC multipoles $P_{\ell, \text{RDCC}}(k)$,
- RDCC FoG damping function $D_{\text{FoG}, \text{RDCC}}(k, \mu)$,
- RDCC small-scale velocity dispersion $\sigma_{v, \text{RDCC}}$,
- relaxation scale k_{rel} and suppression exponent α .

These are stored in a dedicated `rdcc_rsd` structure in `common.h`.

B.3 Input Parameters

The following new input parameters are added to `input.c`:

- `k_rel` — relaxation scale,
- `rdcc_chi_f` — growth-rate suppression,
- `rdcc_chi_delta` — density suppression,
- `rdcc_chi_theta` — velocity divergence suppression,
- `rdcc_chi_v` — velocity dispersion suppression,
- `rdcc_alpha` — suppression exponent.

Default values match the analytical approximations of Appendix A.

B.4 RDCC Growth Rate and Velocity Divergence

CLASS must compute the RDCC growth rate:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (59)$$

The RDCC velocity divergence spectrum is

$$P_{\theta\theta, \text{RDCC}}(k) = P_{\theta\theta}(k) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (60)$$

These must be computed on the same k - z grid used for standard RSD.

B.5 RDCC Density–Velocity Cross Spectrum

The cross spectrum is

$$P_{\delta\theta, \text{RDCC}}(k) = P_{\delta\theta}(k) \left[1 - \chi_{\delta\theta} \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (61)$$

CLASS must compute and store this spectrum for all requested k .

B.6 RDCC Anisotropic Power Spectrum

The full RDCC anisotropic power spectrum is

$$P_{s, \text{RDCC}}(k, \mu) = b^2 P_{\delta\delta, \text{RDCC}}(k) + 2b\mu^2 P_{\delta\theta, \text{RDCC}}(k) + \mu^4 P_{\theta\theta, \text{RDCC}}(k). \quad (62)$$

CLASS must:

- compute $P_{s, \text{RDCC}}(k, \mu)$ on a 2D grid,
- store the result in `rdcc_rsd`,
- make it available to `output.c`.

B.7 RDCC FoG Damping

The RDCC FoG damping function is

$$D_{\text{FoG}, \text{RDCC}}(k, \mu) = \exp \left[- (k\mu\sigma_v)^2 \left(1 - \chi_v \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right) \right]. \quad (63)$$

CLASS must apply this damping to $P_{s, \text{RDCC}}(k, \mu)$.

B.8 RDCC Multipoles

The RDCC multipoles are

$$P_{\ell, \text{RDCC}}(k) = \frac{2\ell + 1}{2} \int_{-1}^1 d\mu P_{s, \text{RDCC}}^{\text{FoG}}(k, \mu) \mathcal{L}_\ell(\mu). \quad (64)$$

CLASS must compute:

- monopole $P_{0, \text{RDCC}}(k)$,
- quadrupole $P_{2, \text{RDCC}}(k)$,
- hexadecapole $P_{4, \text{RDCC}}(k)$.

B.9 Numerical Stability

To ensure numerical stability:

- use logarithmic k -grids for RSD spectra,
- spline-interpolate all RDCC suppression factors,
- precompute scale-dependent factors involving k_{rel} ,
- avoid evaluating suppression factors at extremely small scales,
- ensure smooth transitions between linear and non-linear regimes.

The additional computational cost is modest.

B.10 Summary

This appendix provides a complete implementation strategy for RDCC-modified RSD in CLASS. The modifications are minimal, the numerical cost is low, and the resulting outputs are fully consistent with the analytical and semi-analytic results derived in the main text.

Appendix C: Forecast Tables for DESI, Euclid, Roman, and SKA

This appendix provides forecast tables for detecting the RDCC-modified redshift-space distortion (RSD) signatures with DESI, Euclid, Roman, and SKA. We summarize the expected signal-to-noise ratios, detection significances, and constraints on the relaxation scale k_{rel} and the small-scale velocity dispersion $\sigma_{v,\text{RDCC}}$.

C.1 DESI Forecasts

Observable	RDCC Signature	DESI Sensitivity
$P_2(k)$ small-scale suppression	10–20%	$> 4\sigma$
$P_4(k)$ turnover	at k_{rel}	$> 3\sigma$
Velocity dispersion σ_v	–15–25%	$> 3\sigma$
Growth rate $f_{\text{RDCC}}(k)$	scale-dependent	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 1: DESI forecast sensitivities to RDCC RSD signatures.

C.2 Euclid Forecasts

Observable	RDCC Signature	Euclid Sensitivity
Large-scale $P_2(k)$ enhancement	5–10%	$> 3\sigma$
Small-scale $P_4(k)$ suppression	20–30%	$> 4\sigma$
Turnover in multipoles	at k_{rel}	$> 3\sigma$
Velocity dispersion σ_v	–20%	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.04 h \text{ Mpc}^{-1}$

Table 2: Euclid forecast sensitivities to RDCC RSD signatures.

C.3 Roman Forecasts

Observable	RDCC Signature	Roman Sensitivity
Small-scale $P_4(k)$ suppression	25–35%	$> 5\sigma$
Turnover in $P_2(k)$	at k_{rel}	$> 4\sigma$
Velocity dispersion σ_v	–20–30%	$> 4\sigma$
Large-scale coherent flows	enhanced	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 3: Roman forecast sensitivities to RDCC RSD signatures.

C.4 SKA Forecasts

Observable	RDCC Signature	SKA Sensitivity
Large-scale $P_2(k)$ smoothing	5–15%	$> 5\sigma$
Turnover in $P_4(k)$	at k_{rel}	$> 4\sigma$
Velocity dispersion σ_v	–20%	$> 3\sigma$
Scale-dependent growth rate	detectable	$> 3\sigma$
Constraint on k_{rel}	—	$\pm 0.03 h \text{ Mpc}^{-1}$

Table 4: SKA forecast sensitivities to RDCC RSD signatures.

C.5 Combined Constraints

Parameter	RDCC Prediction	Combined Sensitivity
Relaxation scale k_{rel}	turnover in RSD	$\pm 0.02 h \text{ Mpc}^{-1}$
Velocity dispersion suppression	15–30%	$> 6\sigma$
Large-scale coherent flows	enhanced	$> 5\sigma$
Multipole turnover	at k_{rel}	$> 5\sigma$
Scale-dependent growth rate	detectable	$> 4\sigma$

Table 5: Combined constraints from DESI, Euclid, Roman, and SKA.

C.6 Summary

These forecast tables demonstrate that DESI, Euclid, Roman, and SKA provide strong observational sensitivity to the RDCC RSD signatures:

- suppressed small-scale multipoles,
- enhanced large-scale coherent flows,
- reduced small-scale velocity dispersion,
- a characteristic turnover at k_{rel} ,
- scale-dependent growth rate.

Together, these surveys can constrain the relaxation scale to $\Delta k_{\text{rel}} \sim 0.02 h \text{ Mpc}^{-1}$ and detect the RDCC velocity-field signatures at high significance.

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